

Non Right Angled Triangles



Curriculum Ready
ACMMG: 273



www.mathletics.com

NON RIGHT ANGLED TRIANGLES

$\sin \theta$, $\cos \theta$ and $\tan \theta$ are also useful in non-right angled triangles. This unit will show you how to use these trigonometric ratios to find the sizes of angles and length of sides.



Try to answer these questions now, *before* working through the chapter.

I used to think:

How are sides labelled in relation to the labelling angles of a non right angled triangle? Draw and label a triangle.

The "cosine rule" is used when two sides of the triangle and the angle between them is given
What is this rule?



Answer these questions *after* you have worked through the chapter.

But now I think:

How are sides labelled in relation to the labelling angles of a non right angled triangle? Draw and label a triangle.

The "cosine rule" is used when two sides of the triangle and the angle between them is given
What is this rule?



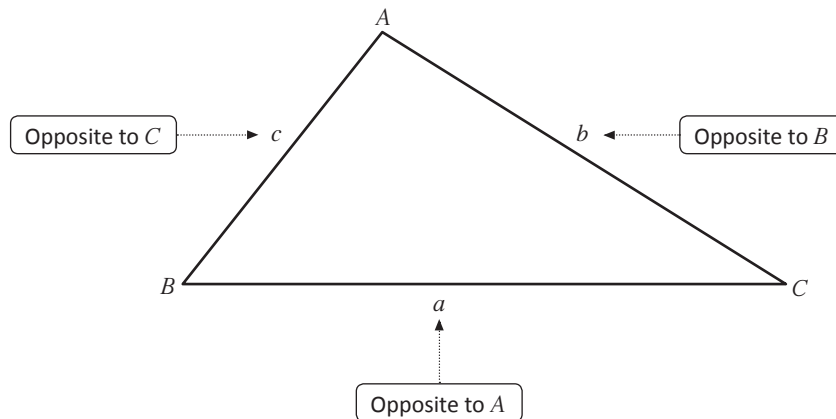
What do I know now that I didn't know before?

Labelling Triangles

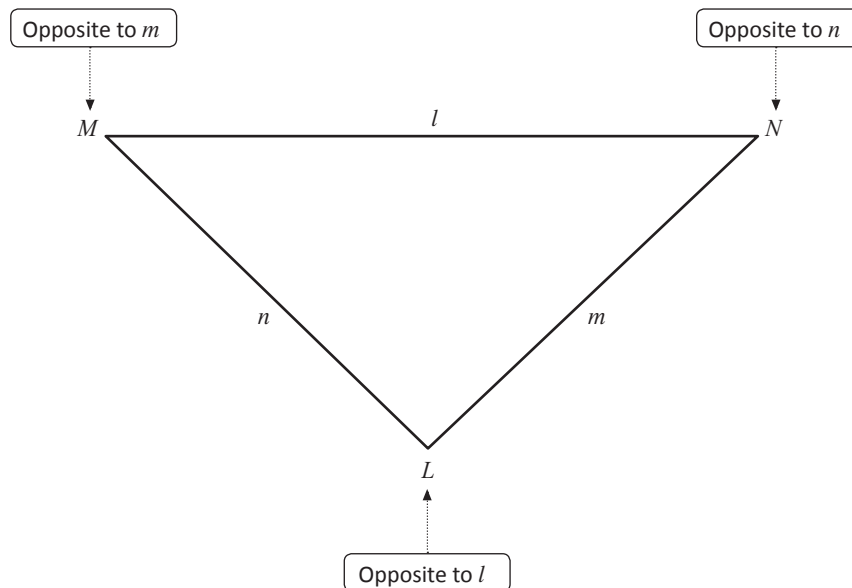
Everyone labels triangles the same way.

Each **angle** is labelled with a **capital** letter, and their opposite **sides** are labelled with the same letter, but **non-capital**.

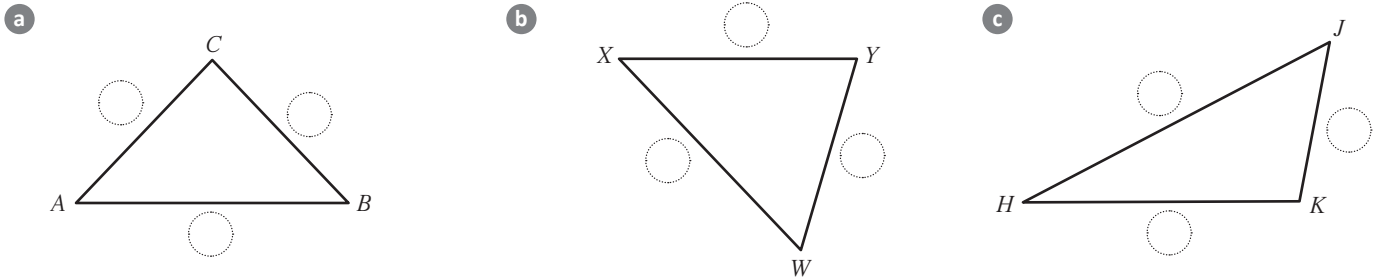
For $\triangle ABC$



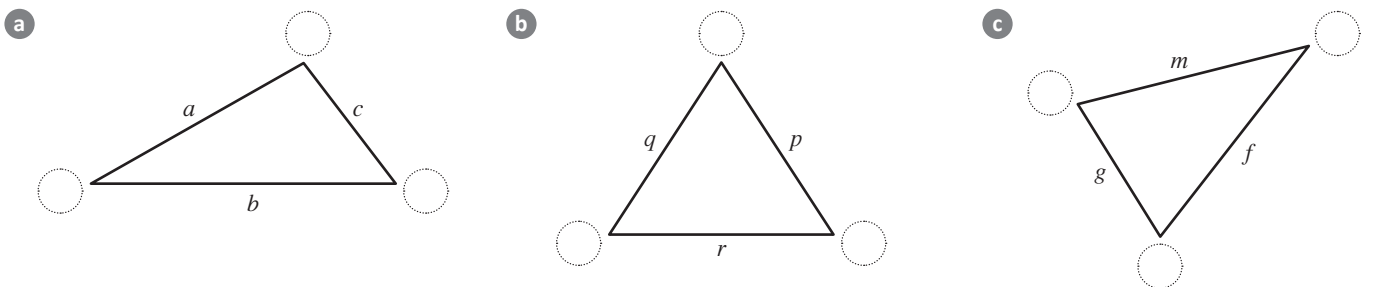
For $\triangle LMN$



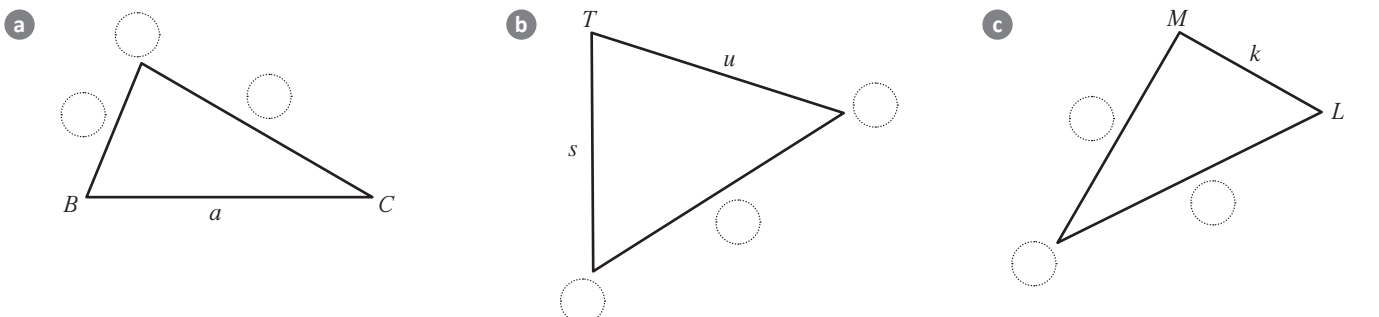
1. Label the sides in the following triangles.



2. Label the angles in the following the triangles.



3. Complete the labelling of the following triangles.



So far we've used trigonometry in right-angled triangles, but we can actually use it to find sides and angles in any triangle!

The Area Rule

$\triangle ABC$ is not right-angled. The perpendicular height, h , is drawn in.

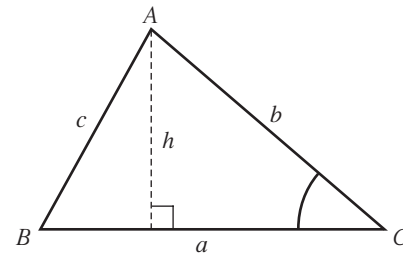
$$\text{Area} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Area} = \frac{1}{2} \times a \times h$$

We can see: $\frac{h}{b} = \sin C$
 $\therefore h = b \sin C$

$$\therefore \text{Area} = \frac{1}{2} \times a \times \overbrace{b \sin C}$$

$$\therefore \boxed{\text{Area} = \frac{1}{2} ab \sin C}$$



So the area is half the product of two sides and the sine of their interior angle. Any two sides and their interior angle will work:

$$\boxed{\text{Area} = \frac{1}{2} ab \sin C}$$

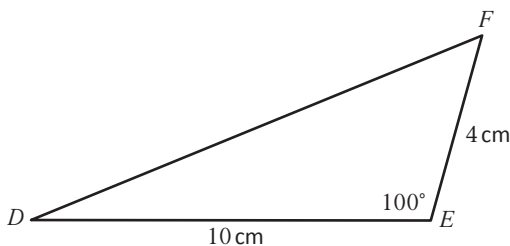
or

$$\boxed{\text{Area} = \frac{1}{2} bc \sin A}$$

or

$$\boxed{\text{Area} = \frac{1}{2} ac \sin B}$$

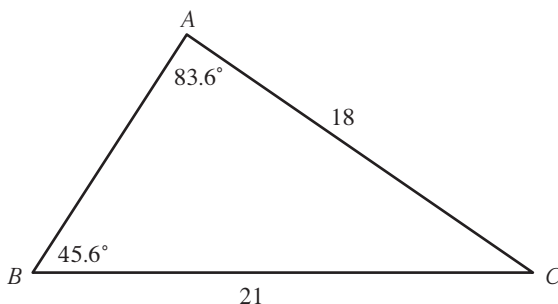
Find the area of $\triangle DEF$ to 1 decimal place



d and f are given, so we will use $\text{Area} = \frac{1}{2} df \sin E$

$$\begin{aligned} \text{Area } \triangle DEF &= \frac{1}{2} df \sin E \\ &= \frac{1}{2} (4)(10) \sin 100^\circ \\ &= 19.696... \approx 19.7 \text{ cm}^2 \text{ (1 decimal place)} \end{aligned}$$

Find the area of $\triangle ABC$ to 1 decimal place



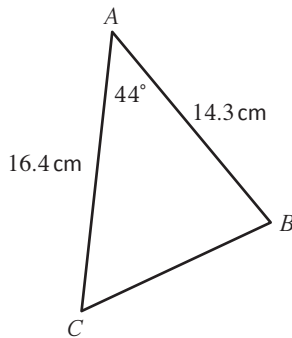
a and b are given, so we will use $\text{Area} = \frac{1}{2} ab \sin C$

$$\angle C = 180^\circ - 83.6^\circ - 45.6^\circ = 50.8^\circ \quad (\text{write this in } \triangle ABC)$$

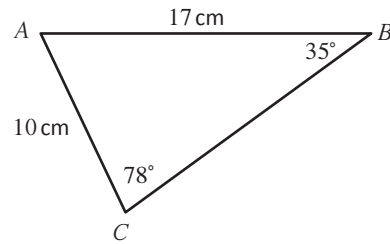
$$\begin{aligned} \text{Area } \triangle ABC &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} (21)(18) \sin 50.8^\circ \\ &= 116.36.. \approx 116.4 \text{ cm}^2 \text{ (1 decimal place)} \end{aligned}$$

1. Find the area of the following triangles to 1 decimal place.

a



b



2. $\triangle ABC$ has $\angle B = 103.3^\circ$, $\angle C = 33.7^\circ$, $b = 21.4\text{ cm}$ and $c = 12.2\text{ cm}$.

a

Draw the triangle.

b

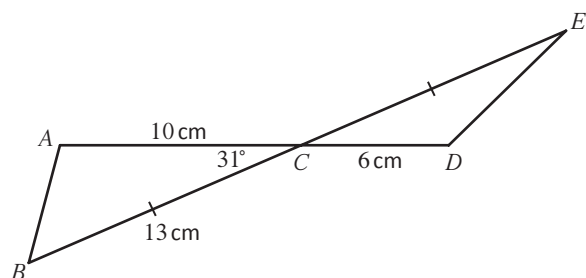
Find $\angle A$ to nearest degree.

c

Find the area of the triangle to 1 decimal place.

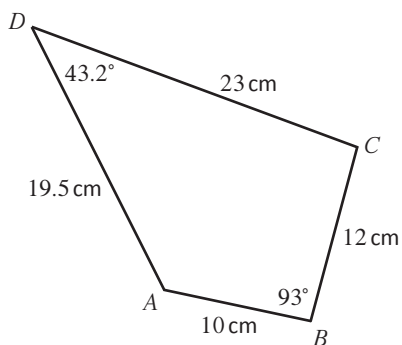
3. Can you find the areas of the following shapes to the nearest square cm?

a

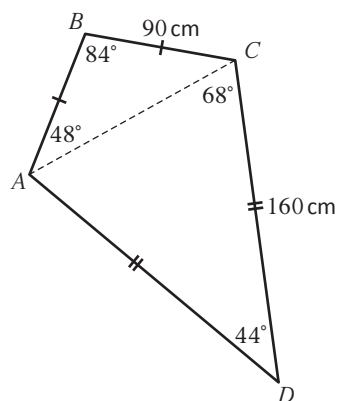


b

Hint: Draw in Diagonal AC

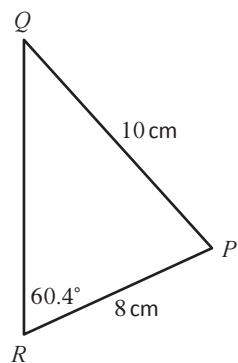


4. Luke wants to make a kite in the shape below. How many square cm of material will he need?

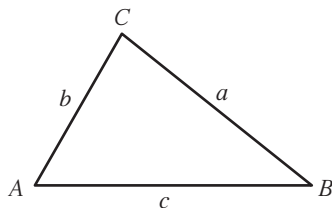


5. The area of the triangle $\triangle PQR$ is 38.6 cm^2 . Find $\angle Q$.

(Hint: Find $\angle P$ first)



The Sine Rule



In any triangle $\triangle ABC$ we have the formulas:

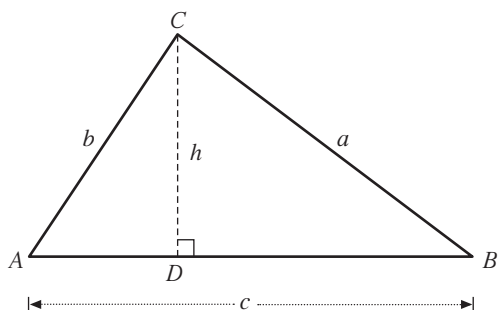
$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

or

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The proof is different depending on whether the triangle is acute or obtuse.

If $\triangle ABC$ is an acute triangle



$$\begin{aligned} \text{In } \triangle ACD: \quad \frac{h}{b} &= \sin A \\ \therefore h &= b \sin A \end{aligned}$$

$$\begin{aligned} \text{In } \triangle BCD: \quad \frac{h}{a} &= \sin B \\ \therefore h &= a \sin B \end{aligned}$$

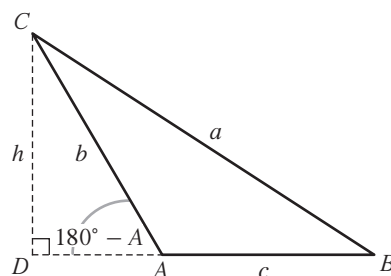
$$\therefore b \sin A = a \sin B \quad (\text{Both equal } h)$$

$$\therefore \frac{\sin A}{a} = \frac{\sin B}{b}$$

If we drew the altitude from B to AC then we could show

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

If $\triangle ABC$ is an obtuse triangle



$$\begin{aligned} \text{In } \triangle BCD: \quad \frac{h}{a} &= \sin B \\ \therefore h &= a \sin B \end{aligned}$$

$$\begin{aligned} \text{In } \triangle ACD: \quad \frac{h}{b} &= \sin(180^\circ - A) \\ \therefore h &= b \sin(180^\circ - A) \\ \therefore h &= b \sin A \end{aligned}$$

Remember
 $\sin(180^\circ - \theta) = \sin \theta$

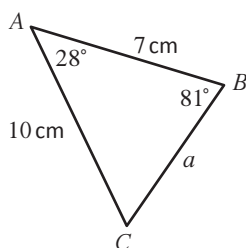
$$\therefore b \sin A = a \sin B \quad (\text{Both equal } h)$$

$$\therefore \frac{\sin A}{a} = \frac{\sin B}{b}$$

If we drew the altitude from B to AC then we could show

$$\frac{\sin A}{a} = \frac{\sin C}{c}$$

Find the value of a in $\triangle ABC$ to 2 decimal places



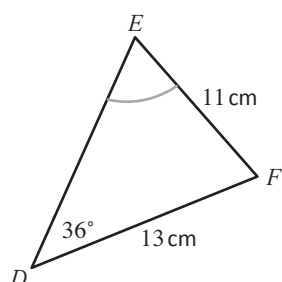
Using the Sine Rule: $\frac{a}{\sin A} = \frac{b}{\sin B}$

$$\therefore \frac{a}{\sin 28^\circ} = \frac{10}{\sin 81^\circ}$$

$$\therefore a = \frac{10 \sin 28^\circ}{\sin 81^\circ}$$

$$a = \frac{10(0.469...)}{0.987...} = 4.753... \approx 4.75 \text{ cm (2 decimal places)}$$

Find the value of $\angle E$ in $\triangle DEF$ to the nearest degree



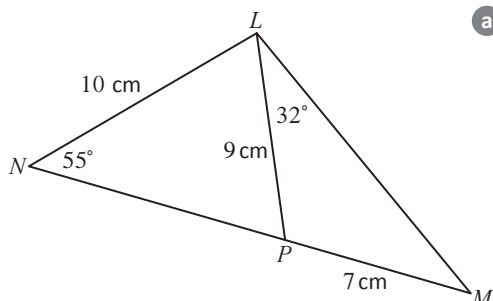
Using the Sine Rule: $\frac{\sin E}{e} = \frac{\sin D}{d}$

$$\therefore \frac{\sin E}{13} = \frac{\sin 36^\circ}{11}$$

$$\therefore \sin E = \frac{13 \sin 36^\circ}{11} = 0.694...$$

$$\therefore E = \sin^{-1}(0.694...) = 43.999...^\circ \approx 44^\circ \text{ (nearest degree)}$$

Use the diagram below to answer these questions



- a Find $\angle LPN$ to the nearest degree.

$$\frac{\sin \angle LPN}{LN} = \frac{\sin \angle LNP}{LP}$$

$$\therefore \frac{\sin \angle LPN}{10} = \frac{\sin 55^\circ}{9}$$

$$\therefore \sin \angle LPN = \frac{10 \sin 55^\circ}{9} = 0.910...$$

$$\therefore \angle LPN = \sin^{-1}(0.910...) = 65.52...^\circ \approx 66^\circ \text{ (nearest degree)}$$

- b Find the length of LM to the nearest cm.

First, find $\angle LPM$

$$\begin{aligned} \angle LPM &= 180^\circ - \angle LPN \\ &= 180^\circ - 66^\circ = 114^\circ \end{aligned}$$

Now, use the sine rule

$$\frac{LM}{\sin \angle LPM} = \frac{PM}{\sin \angle PLM}$$

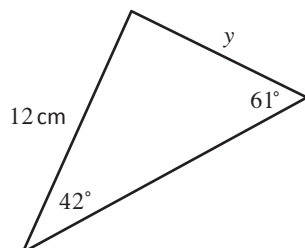
$$\therefore \frac{LM}{\sin 114^\circ} = \frac{7}{\sin 32^\circ}$$

$$\therefore LM = \frac{7 \sin 114^\circ}{\sin 32^\circ} = 12.067... \approx 12 \text{ cm (nearest cm)}$$

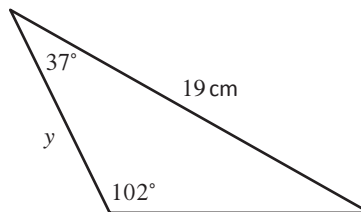
The Sine Rule

1. Find the length y in the triangles below to 1 decimal place.

a

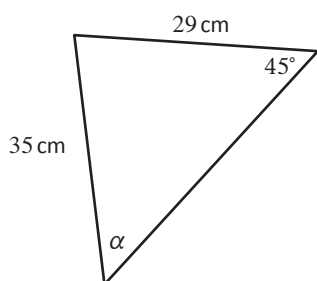


b

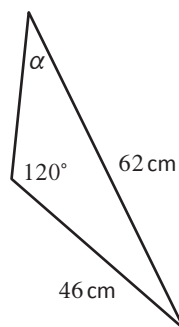


2. Find the size of angle α in the following triangles:

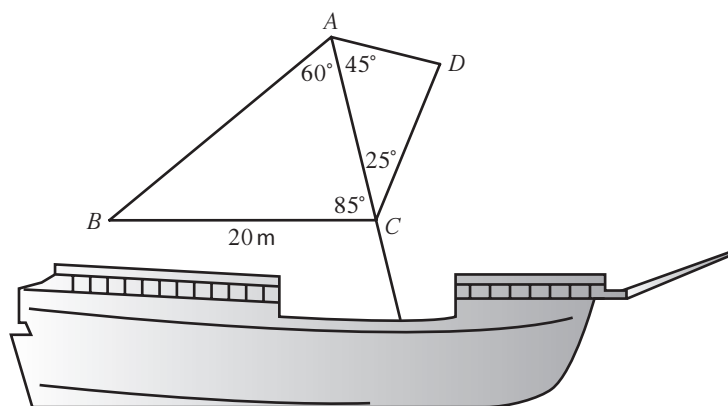
a



b

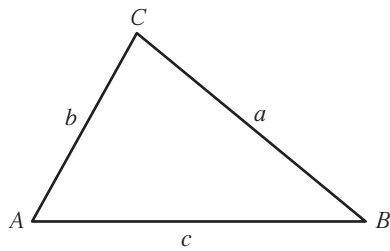


3. A ship has sails as drawn in the diagram below:



- a Find the length of AC .
- b Find the lengths of DC and AD .

The Cosine Rule



In any triangle $\triangle ABC$ we have the formulas

$$a^2 = b^2 + c^2 - 2bc \cos A$$

or

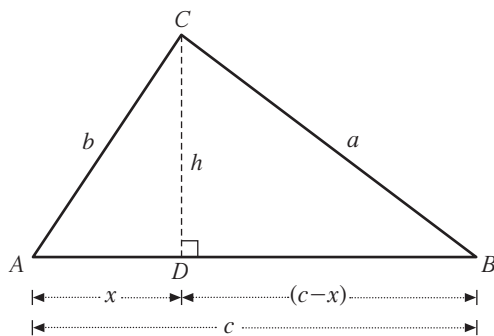
$$b^2 = a^2 + c^2 - 2ac \cos B$$

or

$$c^2 = a^2 + b^2 - 2ab \cos C$$

The proof is different depending on if the triangle is acute or obtuse.

If $\triangle ABC$ is an acute triangle



$$\begin{aligned} \text{In } \triangle CBD: a^2 &= h^2 + (c-x)^2 && \text{(Pythagoras)} \\ &= h^2 + x^2 + c^2 - 2cx \end{aligned}$$

Substitute

Substitute

$$\text{In } \triangle ACD: h^2 + x^2 = b^2 \quad \text{(Pythagoras)}$$

$$\begin{aligned} \text{and } \cos A &= \frac{x}{b} \\ \therefore x &= b \cos A \end{aligned}$$

$$a^2 = b^2 + c^2 - 2cb \cos A$$

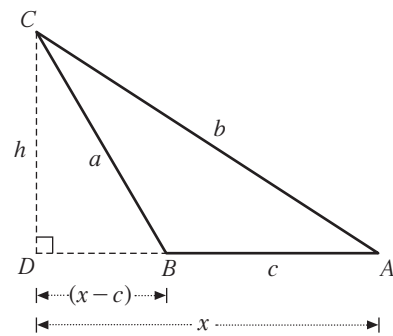
If the h is drawn from other corners we can show

$$b^2 = a^2 + c^2 - 2ac \cos B$$

or

$$c^2 = a^2 + b^2 - 2ab \cos C$$

If $\triangle ABC$ is an obtuse triangle



$$\begin{aligned} \text{In } \triangle CBD: a^2 &= h^2 + (x-c)^2 && \text{(Pythagoras)} \\ &= h^2 + x^2 + c^2 - 2cx \end{aligned}$$

Substitute

Substitute

$$\text{In } \triangle ACD: h^2 + x^2 = b^2 \quad \text{(Pythagoras)}$$

$$\begin{aligned} \text{and } \cos A &= \frac{x}{b} \\ \therefore x &= b \cos A \end{aligned}$$

$$a^2 = b^2 + c^2 - 2cb \cos A$$

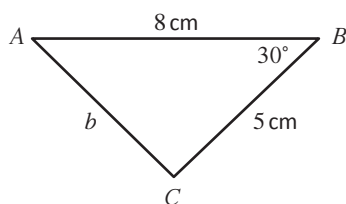
If the h is drawn from other corners we can show

$$b^2 = a^2 + c^2 - 2ac \cos B$$

or

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Find the value of b in $\triangle ABC$ to 2 decimal places



Using the Cosine Rule:

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$\therefore b^2 = 5^2 + 8^2 - 2(5)(8) \cos 30^\circ$$

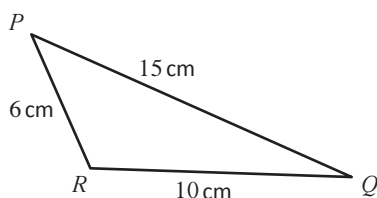
$$\therefore b^2 = 25 + 64 - 80(0.866\dots)$$

$$\therefore b^2 = 19.717\dots$$

$$\therefore b = \sqrt{19.717\dots}$$

$$= 4.440\dots \approx 4.44 \text{ cm (2 decimal places)}$$

Find the value of $\angle Q$ to the nearest degree



Using the Cosine Rule:

$$q^2 = p^2 + r^2 - 2pr \cos Q$$

$$\therefore 6^2 = 10^2 + 15^2 - 2(10)(15) \cos Q$$

Make $\cos Q$ the subject

$$2(10)(15) \cos Q = 10^2 + 15^2 - 6^2$$

$$300 \cos Q = 100 + 225 - 36 = 289$$

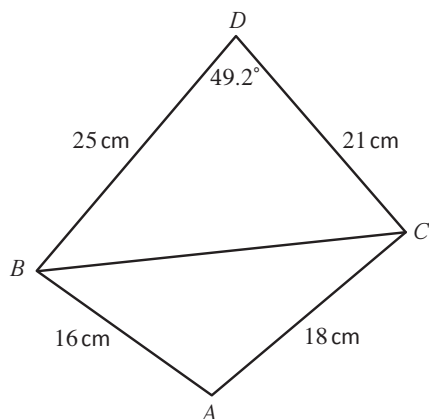
$$\therefore \cos Q = \frac{289}{300} = 0.963\dots$$

Solve for $\angle Q$

$$\angle Q = \cos^{-1}(0.963\dots)$$

$$= 15.56\dots^\circ \approx 16^\circ \text{ (nearest degree)}$$

Find $\angle A$ in the following to 1 decimal place



$$BC^2 = DB^2 + DC^2 - 2 \times DB \times DC \times \cos D$$

This is the cosine rule

$$BC^2 = 25^2 + 21^2 - 2(25)(21) \cos 49.2^\circ$$

$$BC^2 = 379.908\dots$$

$$BC = 19.49\dots \text{ cm}$$

Write this in the diagram

$$BC^2 = AB^2 + AC^2 - 2 \times AB \times AC \times \cos A$$

$$(19.49\dots)^2 = 16^2 + 18^2 - 2(16)(18) \cos A$$

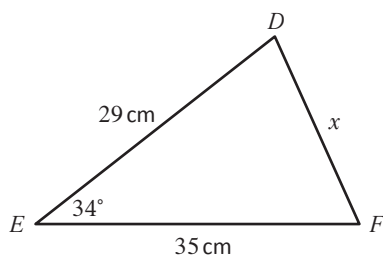
$$\therefore \cos A = \frac{16^2 + 18^2 - (19.49\dots)^2}{2(16)(18)} = 0.347\dots$$

$$\therefore \angle A = \cos^{-1}(0.347\dots) = 69.67\dots^\circ \approx 69.7^\circ \text{ (1 decimal place)}$$

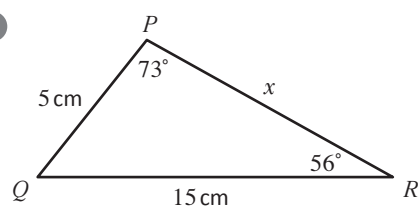
The Cosine Rule

4. Find the length of the missing side x in each of the following to 1 decimal place.

a

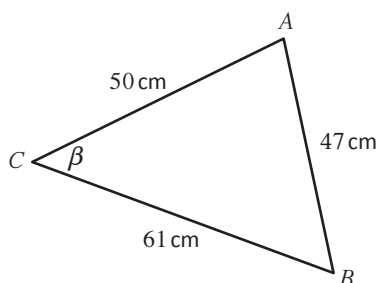


b

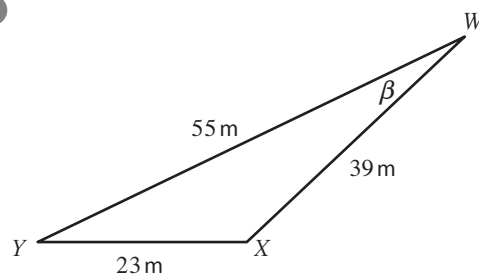


5. Find the value of the angle β in each of the following to the nearest degree.

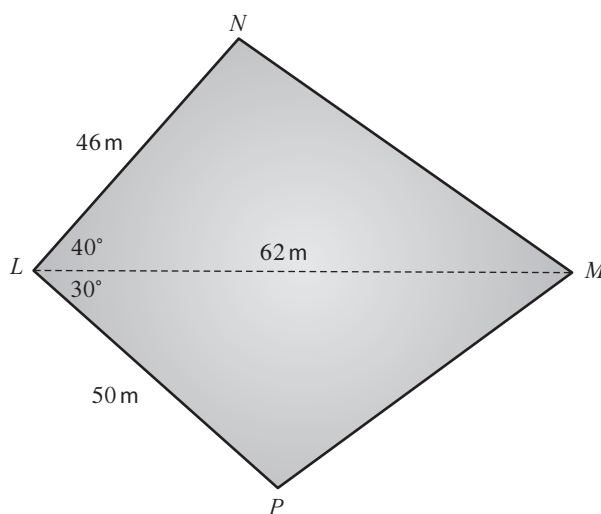
a



b



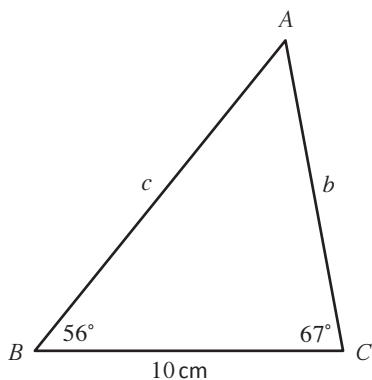
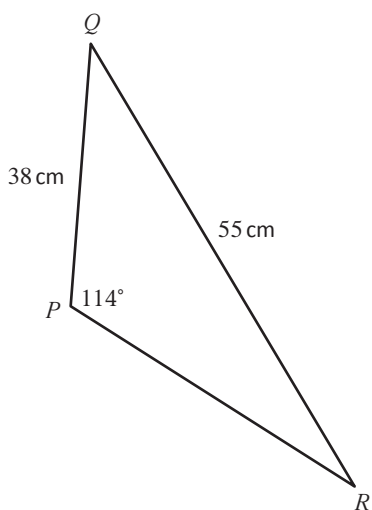
6. A canyon has the following dimensions:



To walk around this canyon, would it be shorter to walk along $LN M$ or $L P M$?
Use working to support your answer.

Both Rules

7. Use the sine and cosine rules to find the missing sides and angles in these triangles to 2 decimal places.

a**b**

An Ambiguous Case For The Sine Rule

Remember that $\sin \theta = \sin(180^\circ - \theta)$. So when the sine rule is used to find an angle (using $\sin^{-1}$) two answers are possible.

For example if $\sin \theta = 0.5$ then $\theta = 30^\circ$ or 150° ($180^\circ - 30^\circ$).

It is our job to determine whether one or both answers are valid. This depends on whether a third angle exists or not.

Find $\angle B$ (to 1 decimal place) in $\triangle ABC$ if $\angle C = 50^\circ$, $c = 14\text{cm}$ and $b = 17.2\text{cm}$

According to the sine rule

$$\begin{aligned}\frac{\sin B}{b} &= \frac{\sin C}{c} \\ \therefore \frac{\sin B}{17.2} &= \frac{\sin 50^\circ}{14} \\ \therefore \sin B &= \frac{17.2 \times \sin 50^\circ}{14} = 0.9411 \\ \therefore \angle B &= \sin^{-1}(0.9411) \\ \therefore \angle B &= 70.2^\circ\end{aligned}$$

or

$$\begin{aligned}\angle B &= 180^\circ - 70.2^\circ \\ &= 109.8^\circ\end{aligned}$$

Add each answer to the given angle. If this sum is less than 180° then the answer is valid because the third angle exists.

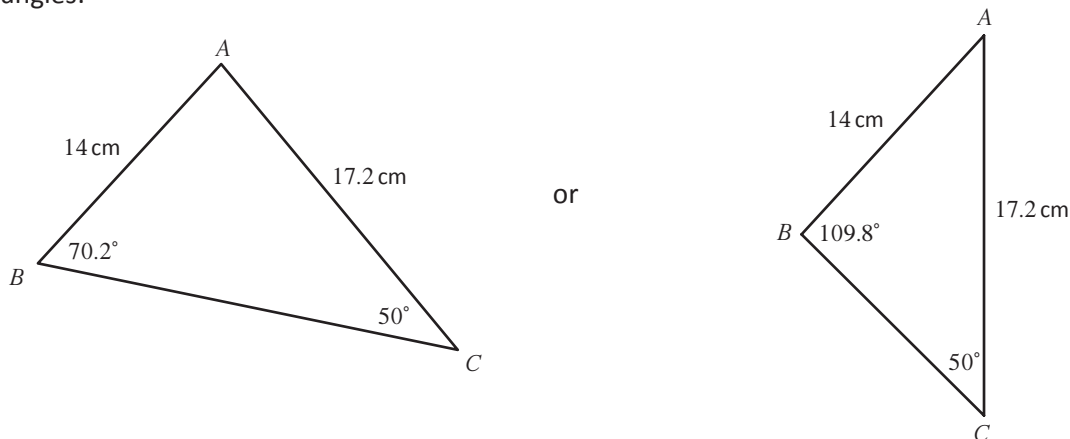
The given angle in the question is $\angle C = 50^\circ$

$$\begin{aligned}\therefore \angle B + \angle C &= 70.2^\circ + 50^\circ \\ &= 120.2^\circ < 180^\circ \\ \therefore \text{valid}\end{aligned}$$

or

$$\begin{aligned}\angle B + \angle C &= 109.8^\circ + 50^\circ \\ &= 159.8^\circ < 180^\circ \\ \therefore \text{valid}\end{aligned}$$

Both sums are less than 180° , and so both answers for $\angle B$ are valid. So $\triangle ABC$ may look like either of these triangles:



A case where both answers are valid (like above) is said to be "ambiguous."

Determining if Answers are Valid

Of the two possible answers, one will **always** be obtuse (90° to 180°) and the other will **always** be acute (0° to 90°).

- So if you know the triangle is acute, then the obtuse angle is invalid.
- If you know the missing angle is obtuse, then the acute angle is invalid.
- The sum of angles in a triangle is 180° . So if the sum of a possible answer and the given angle is 180° or more it means that the third angle in the triangle is 0° or negative. This is impossible, so the answer would be invalid.

Find $\angle L$ (to 1 decimal place) in $\triangle LMN$ if $\angle N = 80^\circ$, $n = 48.6$ cm and $l = 40.5$ cm

According to the sine rule

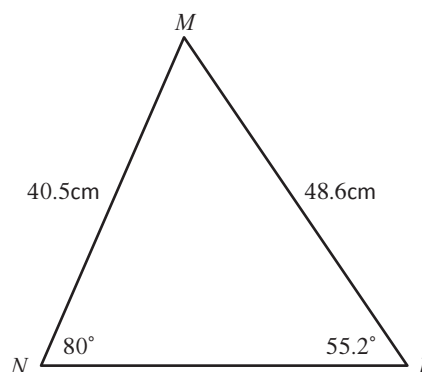
$$\begin{aligned}\frac{\sin L}{l} &= \frac{\sin N}{n} \\ \therefore \frac{\sin L}{40.5} &= \frac{\sin 80^\circ}{48.6} \\ \therefore \sin L &= \frac{40.5 \times \sin 80^\circ}{48.6} = 0.8206... \\ \therefore \angle L &= \sin^{-1}(0.8206...) \\ \therefore \angle L &= 55.1522...^\circ \approx 55.2^\circ \quad \text{or} \quad \angle L = 180^\circ - 55.1522 \\ &= 124.8477 \approx 124.8^\circ\end{aligned}$$

Add each answer to the given angle ($\angle N = 80^\circ$).

$$\begin{aligned}\therefore \angle L + \angle N &= 55.2^\circ + 80^\circ & \text{or} & \quad \angle L + \angle N = 124.8^\circ + 80^\circ \\ &= 135.2^\circ < 180^\circ & & \quad = 204.8^\circ > 180^\circ \\ \therefore \text{valid} & & & \quad \therefore \text{invalid}\end{aligned}$$

So only the **first answer** is valid and $\angle L = 55.2^\circ$. This is **not** an ambiguous case.

$\angle L = 55.2^\circ$ and $\triangle LMN$ looks like this



2. In $\triangle PQR$, $\angle R = 40^\circ$, $p = 20.3\text{cm}$ and $r = 13.9$.

- a Find the two possible values for $\angle P$ to the nearest degree.
- b Is this an ambiguous case (are both angles valid)?
- c Which answer is valid if $\triangle PQR$ is acute? Which answer is invalid?
- d Which answer is valid if $\angle P$ is obtuse? Which answer is invalid?

3. In triangle $\triangle JKL$, $\angle K = 70^\circ$, $k = 51.4\text{cm}$ and $l = 42\text{cm}$.

a Find the valid size(s) of $\angle L$.

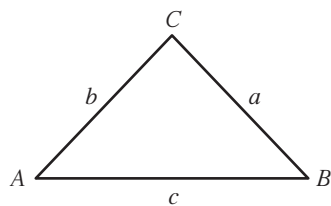
b Find $\angle J$.

c Find the length of j .

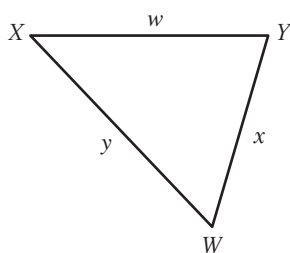
d Draw $\triangle JKL$.

Basics:

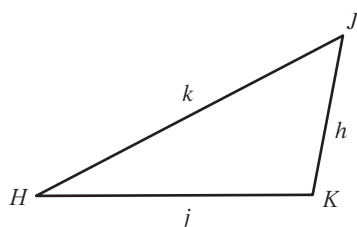
1. a



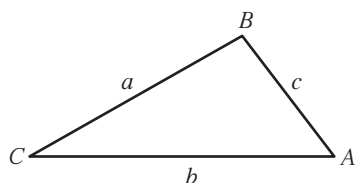
b



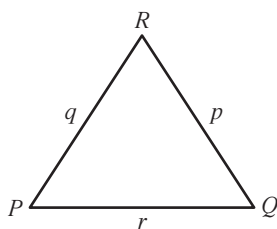
c



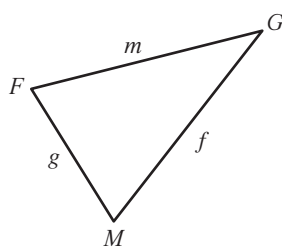
2. a



b

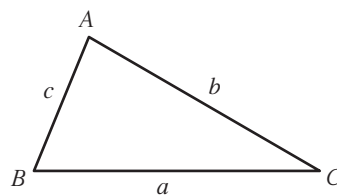


c

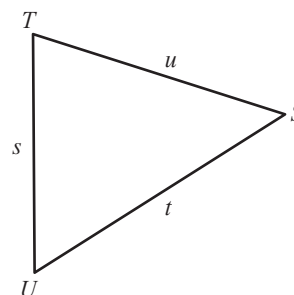


Basics:

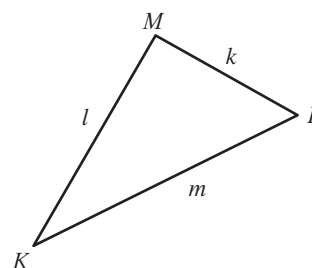
3. a



b



c

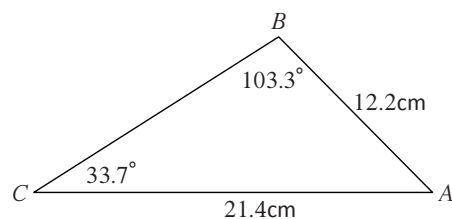


Knowing More:

1. a Area $\triangle ABC = 81.5\text{cm}^2$

b Area $\triangle ABC = 74.2\text{cm}^2$

2. a



b $\angle A = 43^\circ$

c Area $\triangle ABC = 89.0\text{cm}^2$

Knowing More:

3. **a** Total area = $53.563 \approx 54\text{cm}^2$

b Total area = 213cm^2

4. Total area = 12919cm^2

5. $\angle Q = 44.8^\circ$

Using Our Knowledge:

1. **a** $y = 9.2\text{cm}$

b $y = 12.7\text{cm}$

2. **a** $\alpha = 35.9^\circ$

b $\alpha = 40^\circ$

3. **a** $AC = 13.2\text{m}$ (to 1 d.p.)

b $CD \approx 10.0\text{m}$ and $AD \approx 6.0\text{m}$

4. **a** $x = 19.6\text{cm}$ (to 1 d.p.)

b $x = 12.5\text{cm}$ (to 1 d.p.)

5. **a** $\beta = 49^\circ$ (to nearest degree)

b $\beta = 21^\circ$ (to nearest degree)

6. It would be shorter to walk along LPM than LNM

7. **a** $c = 10.99\text{cm}$ (to 2 d.p.)

$b = 9.89\text{cm}$ (to 2 d.p.)

b $q = 27.20\text{cm}$ (to 2 d.p.)

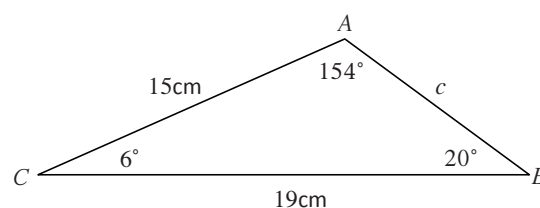
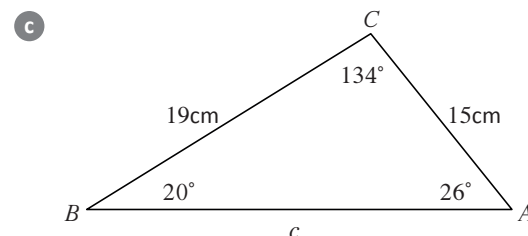
$\angle Q = 26.86^\circ$ (to 2 d.p.)

$\angle R = 39.14^\circ$ (to 2 d.p.)

Thinking More:

1. **a** $\angle A = 26^\circ$ or $\angle A = 154^\circ$

b Both answers are valid; therefore this is an ambiguous case.



d $\angle C = 134^\circ$ or $\angle C = 6^\circ$

2. **a** $\angle C = 70^\circ$ or $\angle C = 110^\circ$

b Both answers are valid; therefore this is an ambiguous case.

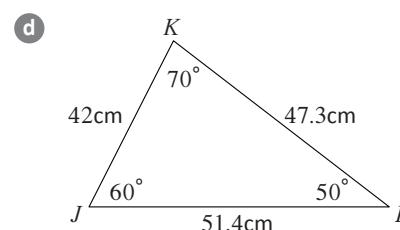
c If $\triangle PQR$ is acute, the angles are in the range $0^\circ - 90^\circ$, $\angle P = 70^\circ$ is the valid answer and $\angle P = 110^\circ$ is invalid.

d If $\triangle PQR$ is obtuse, the angles are in the range $90^\circ - 180^\circ$, $\angle P = 110^\circ$ is the valid answer and $\angle P = 70^\circ$ is invalid.

3. **a** $\angle L = 50^\circ$

b $\angle J = 60^\circ$

c $j = 47.3\text{cm}$





www.mathletics.com